

Lab 01

Software Verification and Validation Lab

SWE 4806



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Lab Title

Modeling Family Relationships using Alloy

Objective

The objective of this lab is to introduce students to relational modeling and formal verification using Alloy. Students will learn how to:

- Model real-world systems using Alloy signatures and relations
- Define constraints using facts
- Write predicates, functions, and assertions
- Use Alloy Analyzer to detect counterexamples
- Understand how local constraints may still allow unexpected global behavior

Topics Required Before This Lab

1. Signatures (sig)

A signature defines a set of objects.

```
1 sig Person {}
```

This defines a set called **Person**.

2. Relations

Relations represent associations between objects.

```
1 sig Person {  
2   father: lone Man  
3 }
```

This means every person may have at most one father.

3. Multiplicity Keywords

Keyword	Meaning
one	Exactly one
lone	Zero or one
some	One or more
set	Any number



4. Facts

Facts define constraints that are always true.

```
1 fact {  
2     no p: Person | p in p.father  
3 }
```

This states that nobody can be their own father.

5. Predicates

Predicates define reusable logical conditions.

```
1 pred example[p: Person] {  
2     some p.father  
3 }
```

6. Functions

Functions return values or sets.

```
1 fun parents[p: Person]: set Person {  
2     p.father + p.mother  
3 }
```

7. Assertions

Assertions are properties we want to verify.

```
1 assert noSelfParent {  
2     no p: Person | p.father = p  
3 }
```

Assertions are checked using the `check` command.

8. Run and Check

Run is used to generate example instances.

```
1 run example for 5
```

Check is used to verify assertions.

```
1 check noSelfParent for 5
```

Relational Operators in Alloy

Union (+)

Combines relations.

```
1 father + mother
```



Transpose ($\sim$)

Reverses a relation.

```
1 wife = ~husband
```

Transitive Closure ($\wedge$)

Used to traverse recursive relationships.

```
1 p.^father
```

This represents all ancestors reachable through the father relation.

System Description

In this lab, students will model a family relationship system with:

- Men and women
- Parent relationships
- Marriage relationships
- Ancestor constraints

The system must satisfy several logical constraints:

- No person can be their own ancestor
- Marriage relationships must be consistent
- No person should marry an ancestor

Naming Convention

Name your files: id_lab02.asl